A Quaternionic Manifold Kernel for Prime Mapping and Multidimensional Density Walks via Dynamical Quantum Phase Transitions

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Abstract

This memorandum presents the architectural framework for a parallelized C++ engineering kernel designed to map discrete prime distributions onto a four-dimensional quaternionic manifold. By leveraging recent breakthroughs in quantum phase processors, the kernel models non-trivial Riemann Zeta zeros as dynamical quantum phase transitions (DQPTs) using simulated Loschmidt amplitudes. This methodology transforms multidimensional density walks from stochastic processes into rigid, self-adjoint chaotic operator flows. The system continuously monitors algebraic rifts, non-commutativity bounds, and invariant structural safety margins. By incorporating formal proofs establishing that over 67% of zeta zeros are simple and structurally locked to the critical line, the framework establishes a rigorous spectral boundary, replacing exponential classical verification heuristics with polynomial-bound quantum-classical emulation.

1. Introduction and Topological Mapping

The spatial distribution of prime numbers remains an foundational intersection between algebraic number theory and mathematical physics. This kernel establishes a parallelized architecture implemented in high-performance C++ to project discrete primes onto a 4D quaternionic manifold. By utilizing spatial non-commutativity, the mapping exhibits localized algebraic constraints that bound density variations and track structural deformations (rifts) in real-time.

2. Dynamical Quantum Phase Transitions

Rather than treating Riemann Zeta zeros as static complex coordinates, the engine models their positions using dynamic complex phasors linked to simulated Loschmidt amplitudes. Zeros are mapped directly onto sudden alterations in the system's rate of change, formalizing the framework within Dynamical Quantum Phase Transitions (DQPTs). This formulation shifts the problem into a spectral-reality paradigm, aligning the zeros with real quadratic energy levels.

3. Operator Stability and Structural Safety

The architectural stability of the manifold relies on the chaotic operator flow governed by quantum-like chaos matrices. By constraining the spacing intervals of the projected primes with formal parameters (>67% simple zeros bound on the critical line), the engine prevents localized numerical divergence. Invariant structural safety is preserved across parallel computation threads by enforcing strict, self-adjoint spectral boundaries during extended multidimensional density walks.